

Brandon Ng

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EDUCATION

B.S Data Science in Machine Learning and AI
University of California, San Diego

Expected Graduation: June 2026

GPA: 3.8

Relevant Coursework: Principles of Data Science, Statistical Methods, Practice of Data Science, Exploratory Data Analysis and Inference, Intro. Data Management, Discrete Mathematics, Vector Calculus, Linear Algebra, Data Structures and Algorithms, Computer Architecture, Probability and Statistics, Theoretical Foundations of Data Science I & II, Intro to Data Mining, Probabilistic Modeling and ML, Systems for Scalable Analytics, Representation Learning, Deep Learning/Natural Language

EXPERIENCES

Comcast | Software Engineer Intern

June 2025 - Aug 2025

- Refactored and optimized a large-scale recommendation model, migrating from TensorFlow to PyTorch to enable a direct comparison of training efficiency, data pipelines, and library performance for scalable machine learning deployment
- Engineered preprocessing and distributed training pipelines, integrating Hugging Face and Petastorm with PyTorch DataLoaders, implementing a custom sampled softmax loss, and leveraging TorchDistributor for multi-node training
- Conducted A/B benchmarking of PyTorch vs. production TensorFlow models over a 7-day test, achieving higher relevance scores, faster training times, and improved content recommendations, while observing slightly narrower coverage

California Department of Public Health | Software Engineer Student Intern

Feb 2025 - May 2025

- Revamped R setup and RecordLinkage workflows, adding sample datasets and improved RStudio guidance, while researching RecordLinkage string comparison algorithms and weighting schemes to enhance duplicate detection and data quality
- Digitized and deployed the Lead Poisoning Follow-Up Form, transforming a legacy PDF into an interactive Power App, designing a structured SharePoint schema, and creating an end-to-end data pipeline to streamline data collection
- Spearheaded development of modernized backend workflows and data systems, analyzing RASCAL, migrating on-premise servers to the cloud via REDCap and Azure, and leveraging Power Automate to improve scalability and reduce manual effort

CIP4Gov | Full-Stack Software Engineer Intern

Jun 2024 – Sept 2024

- Developed a dynamic Gantt chart using ApexCharts to visualize and improve clarity of project timelines and task progressions
- Reduced data latency by refining API endpoints and synchronizing Angular services with Django views to streamline data
- Enhanced SQL database by restructuring tables to efficiently track budgets and integrated an Excel export feature for reporting

Tech4Good Lab | Back-End Web Developer Team Lead

Jun 2023 – Sep 2023

- Led a team of 10 backend developers, driving technical growth and success through hands-on support and strategic direction
- Collaborated with cross-functional teams to discuss 10 project designs, ensuring successful integration of 15+ components
- Revamped Firebase schema by creating over 20 new fields and 3 new entities to align with evolving website requirements

Tech4Good Lab | Back-End Web Developer

Jan 2023 – Jun 2023

- Populated over 50 test entities in Cloud Firestore using dispatch functions and data models to validate web features pre-launch
- Built 4 dynamic UIs with RxJS and NgRx functions to asynchronously manage and update user data across multiple webpages
- Streamlined data retrieval with 10+ async and selector functions using Observables, ensuring accurate data pipelines.

PROJECTS

Plant Disease Detector

Python, PyTorch, Pandas, CV

Developed a convolutional neural network using PyTorch to classify over 54,000 leaf images and detect 38 plant diseases across 14 species. Improved model performance by optimizing data pipelines and fine-tuning hyperparameters, achieving 89% accuracy rate

ChatGPT EDA

Python, Pandas, Scikit-learn

Performed exploratory data analysis to assess the impact of AI tools like ChatGPT on UCSD's academic environment. Used statistical tests and visualizations to uncover relationships between study habits, grades, and ChatGPT usage across departments

Culinary Insights

Python, Pandas, Scikit-learn

Analyzed recipes on Food.com and built a Random Forest model to predict recipe healthiness based on eight nutritional features. Enhanced model accuracy through grid search optimization and applied hypothesis testing to identify dietary trends within recipes

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS, Java, C/C++, Swift, Assembly, R, SQL, OOP

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, Tensorflow, TorchDistributor, HuggingFace

Frameworks: Angular.js, React.js, Next.js, Node.js, Firebase, Django, NoSQL

Tools: Git, Github, VSCode, Jupyter Notebook, Docker, Selenium, Postman, AWS S3, Apache Spark, Databricks